

Aaron Kiem

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SUMMARY

Electrical Systems Engineering senior with hands-on project experience in circuit design, PCB layout, and power electronics simulation. Seeking full-time and internship positions in manufacturing, electrical integration, and other engineering industries.

EDUCATION

Arizona State University, Ira A. Fulton School of Engineering

Tempe, AZ

B.S.E. in Electrical Systems Engineering; GPA: 3.75

Expected May 2027

- Relevant Coursework: Design of Electrical Systems, Electrical Fundamentals, Computational Modeling of Engineering Systems, CADD and Solid Modeling

LANGUAGES & TECHNOLOGIES

Design & Simulation: SOLIDWORKS, AutoCAD, KiCAD, LTSpice, MATLAB, Simulink, Simscape

Electrical Systems: Circuit/schematic design, PCB layout, power electronics simulation, multimeter diagnostics

Hardware Integration: Arduino IDE, MPLAB X IDE, ESP32 and PIC18 microcontrollers

Certifications (MathWorks): Power Electronics Simulation, Circuit Simulation, MATLAB, Simulink, Simscape

PROJECTS

Robotic Hand for Machining Operations | Senior Capstone, Los Alamos National Lab

Fall 2026 – Spring 2027

- Designing a robotically actuated gripper and machine-tending system for tool exchange, part handling, and milling inside a sealed glovebox
- Developing fixtures and an operator-controlled arm to reduce manual handling of sharp tooling on a Haas TL-1 lathe

PCB Metal Detection System | Electrical Design Coordinator, 6-person team

Spring 2026

- Designed the power and sensor PCB in KiCAD, integrating ESP32 and PIC18 microcontrollers for metal detection
- Modeled 3D printed mounts in SOLIDWORKS and documented design decisions in a GitHub report
- Managed the team schedule and budget, coordinating deliverables across 6 engineers

Clap-Activated Lighting Control System | Circuit Project Manager, 4-person team

Fall 2025

- Built a 4-subsystem sensor/actuator network with a shared hub, integrating all subsystems into one control system
- Drove sound-triggered, touchless, and daylight-based LED responses by programming a PIC18F47K42 microcontroller circuit

Five-Box Coded Lock System | Team Lead, 4-person team

Spring 2025

- Led a 4-person team to design and 3D print 5 independently keyed lockboxes run from a single Arduino UNO and shared keypad
- Engineered motor-driven locking mechanisms routed through one central microcontroller circuit

EXPERIENCE

Shift Manager

Sep 2022 – Present

Pizza Hut

Queen Creek, AZ

- Manage opening/closing operations, cash handling, and inventory across 5 locations; ServSafe certified

Delivery Driver

May 2025 – Present

DoorDash

Queen Creek – Tempe, AZ

- Independently manage delivery logistics and time-sensitive routing whilst being physically adept

ACTIVITIES

Blender & 3D Modeling: Independent projects in 3D modeling, rigging, procedural node texturing, VFX tracking, and LiDAR scanning texturing, VFX tracking, and LiDAR scanning

3D Printing: Personal print practice using SOLIDWORKS and Sketchfab designs in Ultimaker Cura

Game Design: Independent projects applying AI-assisted coding and troubleshooting